



The malleability of sampling's impact on evaluation: Sampling goals moderate the evaluative impact of sampling a stimulus[☆]

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ABSTRACT

People often have some degree of choice over the stimuli they sample and learn more about. These sampling decisions can play an important role in evaluative learning, with recent work showing that sampling a stimulus more frequently predicts a positive shift in its evaluation (Hütter, Niese, & Ihmels, 2022). The current work suggests sampling does not merely have a direct effect of positivity on evaluations, but instead is malleable to people's interpretations of their sampling behavior's meaning. Across five experiments, participants sampled faces to interact with across a series of trials. On each trial, the sampled face was paired with a positive or negative image, and we manipulated participants' sampling goals. Participants given a goal to sample for positivity showed a positive evaluative shift toward the faces they sampled more frequently, regardless of whether it was consistently paired with positive or negative images. Participants given a goal to sample in a balanced way tended to show a similar but weaker effect on evaluative shift. Finally, this effect was eliminated (or reversed) among participants given a goal to sample for negativity. Complementary shifts in evaluation were also observed for faces participants chose not to sample. Thus, these results highlight the role of people's interpretations of what their sampling behavior denotes: in contexts in which sampling should not necessarily predict liking (i.e., when one's goal is to sample for negativity), sampling a stimulus more (vs. less) often does not create positive evaluative shifts.

Much of daily life is characterized by at least some degree of autonomy: we regularly make decisions about the people, situations, objects, etc. that we choose to learn more about (or not). In this way, attitudes are not developed merely via passive exposure to information about various attitude objects, but rather, are often shaped by people's active sampling decisions (Ariely & Norton, 2008; Denrell, 2005). Thus, a true understanding of how evaluative learning processes unfold in real life requires also understanding how the ways in which people receive that information (e.g., via the autonomous decision to sample it) shapes the process.

However, this integral feature of attitude learning in real life is omitted by most research on evaluative learning, which typically employs paradigms that strip participants of autonomy and require them to passively learn about attitude objects within perfectly balanced stimulus distributions. For instance, even within research exploring the consequences of approach behavior, participants are often directly told which stimuli to approach and which to avoid (e.g., Cacioppo, Priester, & Berntson, 1993; Hütter & Genschow, 2020; Van Dessel, De Houwer,

Gast, & Smith, 2015; Van Dessel, Eder, & Hughes, 2018). There is undoubtedly value to such procedures in that they provide high experimental control. However, if our ultimate goal is to understand how evaluative learning unfolds in real life, it is critical that we supplement these findings with other work incorporating features typical of learning processes in everyday life (e.g., autonomy).

In the current work, we employ a paradigm that provides participants with the autonomy to decide how often they sample various stimuli over a series of trials while having different sampling goals. In doing so, we not only replicate recent findings showing that the decision to sample a stimulus itself predicts a positive shift in evaluation (Hütter et al., 2022), but find evidence that this effect is malleable to people's understanding of what their sampling behavior denotes in a given context. In this way, the current work sheds light on the impact of sampling decisions on evaluative learning, providing further insight into how and when they predict a positive shift in people's attitudes.

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1. The Role of Sampling in Evaluative Learning

Attitudes enhance survival by providing a guide based on past learning as to which stimuli are likely to be positive and should be approached and which are likely to be negative and should be avoided (Allport, 1935; Campbell, 1963). Thus, it is not surprising that when people are given autonomy over the stimuli they sample, they tend to sample those that they believe will be positive (Denrell, 2005; Fazio, Eiser, & Shook, 2004). This means that, in contrast to the balanced distributions of positive and negative information typically presented to participants in evaluative learning paradigms (e.g., an equal number of attitude objects paired with positive images as attitude objects paired with negative stimuli), the distributions people create for themselves when given autonomy are typically primarily positive (e.g., Hütter et al., 2022). Additionally, this behavior tends to create asymmetries in learning as people sample stimuli they incorrectly believe are positive, giving them the opportunity to recognize and correct these errors, but avoid sampling stimuli they incorrectly believe are negative, preventing them from even recognizing these errors in their evaluations (Denrell, 2005; Fazio et al., 2004).

In addition to these valence asymmetries in learning, recent research demonstrates that the sampling decision itself can also confer valence (Hütter et al., 2022). This work provided participants the opportunity to sample from a set of stimuli which were then paired with positive or negative images. Beyond the typical evaluative conditioning (EC) effect in which stimuli paired with positive images were liked more than stimuli paired with negative images, there was also a main effect of sampling such that the more frequently a stimulus was sampled, the more positively it was rated afterward. This effect held even among those stimuli that were consistently paired with negative images. Additionally, a control sample of participants who passively viewed the sequences of pairings created by the autonomous participants did not show evidence of this sampling decision effect in which more frequently presented stimuli are liked more. Instead, the effect of valence pairing tended to get stronger with more samples in the passive control condition. Thus, this work suggests that sampling decisions not only impact evaluative learning by altering the information available to learn from, but can have a direct impact on evaluations as well.

Such an effect is consistent with past research arguing that prior behavior can shape people's attitudes (Bem, 1972; Festinger, 1957). For example, a host of studies demonstrate that the act of approaching a stimulus can lead people to like it more, and avoiding a stimulus can lead people to like it less (e.g., Cacioppo et al., 1993; Hütter & Genschow, 2020; Van Dessel et al., 2015, 2018). Similarly, one study used a modified version of the sampling paradigm described in the previous paragraph in which participants were only allowed to choose to sample half of the stimuli; this study found that the stimuli participants chose *not* to sample were evaluated more negatively afterward (Hütter et al., 2022). That is, not only did people apparently use their behavior about sampling a stimulus as informative that they liked it, but they also used their behavior not to sample a stimulus as informative that they disliked it.

However, there is more than one route through which such effects could occur. In particular, there are at least two ways in which sampling behavior might predict a positive evaluative shift, both stemming from the fact that because people tend to sample for stimuli that have positive consequences (e.g., Denrell, 2005). First, because sampling behaviors typically produce positive outcomes, people may form a direct association between sampling and positivity, causing the action itself to be positive and to confer valence upon sampled objects. Alternatively, it may be that people's interpretation of what their sampling behavior implies is critical. That is, because they typically sample for stimuli that have positive consequences, it may be reasonable to infer most of the

time that if one samples something often, it is more likely to be positive. Such a distinction mirrors work in the literature on approach-avoidance (AA) that first theorized AA effects on evaluation may be driven by a mere association between approach-positivity and avoidance-negativity (e.g., Cacioppo et al., 1993), whereas more recent work suggests it is people's inferences about their AA behavior that is critical for its impact on evaluation (e.g., Corneille & Stahl, 2019; Van Dessel, Hughes, & De Houwer, 2019). Following a similar logic in the current context, this would suggest that rather than sampling behavior always predicting a positive evaluative shift, in contexts for which this inference does not apply (e.g., if someone is sampling for stimuli that have negative consequences), increased sampling should not predict a positive evaluative shift. Thus, an important next step in understanding the sampling decision effect is to test between these two potential processes underlying it.

2. The current research

In the current research, we explore whether people's interpretation of their sampling behavior influences the effect of sampling decisions on evaluation. We employed the original paradigm demonstrating the sampling decision effect, in which participants are given autonomy to repeatedly sample from a set of faces (i.e., the conditioned stimuli, or CSs), which are then paired with either positive or negative images (i.e., the unconditioned stimuli, or USs), and we measure the pre- to post-evaluative shift of the CSs (Hütter et al., 2022). Additionally, each high autonomy participant is yoked to another low autonomy, control participant who passively views the pairings, allowing us to test the role of sampling decisions *per se*. Critically, we modified the paradigm for the purposes of the current experiments by giving high autonomy participants different sampling goals.

In Experiment 1, high autonomy participants were randomly assigned to receive instructions asking them to primarily sample positively-paired or negatively-paired faces. Experiment 2 added an additional condition asking some high autonomy participants to sample a balanced number of positively-paired and negatively-paired faces. If there is merely a direct positive associative influence of sampling decisions on evaluations, then we should expect to replicate the sampling decision effect, whereby sampling a CS more often leads to more liking among high (vs. low) autonomy participants, regardless of sampling instructions. If, instead, interpretation plays a critical role, we should expect the sampling decision effect to be moderated by instruction condition, such that it is strongest when people have the goal to sample positively-paired stimuli and eliminated (or even reversed) when people have the goal to sample negatively-paired stimuli.

Experiments 3 and 4 sought to extend the findings from the first two experiments. Specifically, Experiment 3 sought to test the role of interpretation of the decision *not* to sample stimuli. To do so, Experiment 3 replicated Experiments 1 and 2, but only allowed high autonomy participants to sample half of the CSs. Previous work using this version of the task found a negative evaluative shift for unsampled stimuli (Hütter et al., 2022). If choosing not to sample a stimulus merely shapes evaluations via a direct association with negativity, we should expect to replicate this effect regardless of instruction condition. However, if interpretation plays a critical role, we should expect that the unsampled CSs are evaluated most negatively when people have the goal to sample positively-paired stimuli, and this effect should be eliminated when people have the goal to sample negatively-paired stimuli. Finally, Experiment 4 sought to replicate the effects of the previous experiments using a more naturalistic version of the task with less direct sampling instructions. In this way, Experiment 4 sought to demonstrate how giving people task goals that encourage certain types of sampling (rather than directly instructing people to sample for positivity, negativity, or in

a balanced manner) that mirror different real-world sampling goals can replicate the current effects, and indeed, may even strengthen them.

3. Experiments 1 and 2

Two experiments provided initial pre-registered¹ tests of the malleability of the sampling decision effect by altering people's sampling goals. Specifically, Experiment 1 told participants in the high autonomy condition that they could sample faces however they wanted, but that it would be helpful if they primarily sampled positively (or negatively) paired stimuli. Experiment 2 replicated this design but incorporated a third condition asking participants to sample in a balanced manner. If the sampling decision effect is driven by a direct association with positivity, we should expect more frequently sampled stimuli to be liked more among high (vs. low) autonomy participants regardless of sampling instruction condition. However, if as we hypothesize, the effect is sensitive to people's interpretation of the meaning of their sampling behavior, we should expect a positive influence of sampling decision on evaluation among high autonomy participants given the goal to sample positively-paired stimuli, but not among high autonomy participants given the goal to sample negatively-paired stimuli.

3.1. Method

3.1.1. Participants

While we use multi-level modeling to analyze most of our data in the current experiments, we approximated the required sample size by basing the power analysis on a simpler, mixed-design ANOVA that reconceptualized number of samples as a binary (high vs. low) variable.² Using this approach, 152 participants in Experiment 1 and 180 participants in Experiment 2 would provide adequate power ($1 - \beta = 0.80$) to detect a small effect of size $f = 0.125$ (Faul et al., 2007). We aimed to recruit slightly above these amounts to cope with potential data loss due to technical or human error.

In total, we received 168 participant responses in Experiment 1 and 199 participant responses in Experiment 2 from the recruitment website Prolific (www.prolific.co). We only accepted participants with an approval rate above 90% who were between 18 and 60 years old and indicated English as their first language. The experiment lasted roughly 15 min and participants were paid 1.60£ for their participation. As the low-autonomy condition could only be run after data from some high-autonomy participants was available, we ran conditions in alternating blocks of participants. Twenty-eight responses were excluded in Experiment 1 due to a technical error in the yoking procedure, resulting in 70 valid participant pairs (79 women, 56 men, 3 preferred to self-identify, 1 preferred not to say, $M_{age} = 33.35$, $SD_{age} = 11.18$). Three responses were excluded in Experiment 2 due to a technical error in the yoking procedure, resulting in 98 valid participant pairs (103 women, 92 men, $M_{age} = 36.25$, $SD_{age} = 10.59$). Thus, based on the final sample sizes, sensitivity analyses using the same approach we employed to initially determine sample size suggest the smallest effect size we could detect at 80% power ($\alpha = 0.05$) would be $f = 0.129$ in Experiment 1 and $f = 0.119$ in Experiment 2 (Faul et al. 2007).

3.1.2. Design

Experiment 1 implemented a 2 (paired valence: positive vs. negative) $\times$ 2 (autonomy: high vs. low) $\times$ 2 (sampling instructions: positive vs. negative) mixed design with autonomy and sampling instructions as between-participants factors. Experiment 2 implemented a 2 (paired

valence: positive vs. negative) $\times$ 2 (autonomy: high vs. low) $\times$ 3 (sampling instructions: positive vs. balanced vs. negative) mixed design with autonomy and sampling instructions as between-participants factors. Additionally, the number of samples per CS was introduced as a continuous quasi-experimental predictor in both experiments.

3.1.3. Materials and procedure

3.1.3.1. Selection and pre-evaluations of CSs. The experiments started with participants pre-rating 50 faces (i.e., "What is your impression of this face?") on a continuous rating scale with the endpoints "negative" and "positive," which was transformed to a 200-point scale by the computer program. These ratings served as participants' pre-evaluations, and the 12 most neutral faces (i.e., symmetrically surrounding the median rating) were selected as CSs for the conditioning phase.³

3.1.3.2. Sampling goal manipulation. Participants then read a brief introduction of the learning task, informing them that they would have interactions in which faces would be paired with positive or negative images. High autonomy participants were informed that who they chose to interact with on each trial was up to them. In addition, high autonomy participants were randomly assigned a sampling goal. In the positive sampling instructions condition, participants were asked to engage in primarily positive interactions; in the negative sampling instructions condition, participants were asked to engage in primarily negative interactions. However, we also reminded participants that they could freely choose with whom to interact, including with faces paired with the opposite valence of their goal and that they should, "Simply keep in mind that it would be helpful for our research if you primarily engage in [positive/negative] interactions." Experiment 2 also included a balanced sampling instructions condition. Participants randomly assigned to this condition were asked to try to have roughly an equal amount of positive and negative interactions.

3.1.3.3. Sampling $\times$ evaluative conditioning learning task. Participants in the high-autonomy condition were then presented with an overview of the 12 selected CSs arranged at random in a four-by-three grid. Each participant made 60 sampling decisions, in which they were free to view each of the 12 CSs as often and in any order they wished. To start an interaction, participants simply clicked on one of the faces. After clicking on a face, the display of the 12 faces disappeared, and the selected CS appeared together with a US of its assigned valence. Six of the 12 CSs were always paired with positive USs, while the other six CSs were always paired with negative USs. The assignment of CSs to paired US valence was randomly determined for each participant anew. Each US image was used only once. The conditioning trial lasted 2500 ms, after which participants made their next sampling decision. The number of remaining interactions was shown at the bottom of the screen. The position of the CS in the four-by-three grid was randomized for each participant and fixed across the whole learning face.

For USs, we used two pre-selected pools of positive ($N = 60$) and negative pictures ($N = 70$) from the International Affective Picture System (Lang et al., 2008). The two sets were selected so that their mean standard values on valence were equally distant from the neutral

¹ Links to the pre-registrations for all experiments are reported in the Open Practices section at the end of the article.

² More specific details about the power analyses are included in the pre-registrations for each experiment.

³ Due to a coding error, participants in the yoked condition of Experiment 1 all received only the same twelve faces to rate and view as part of the study. Thus, while some caution should be made when making comparisons across high vs. low autonomy groups in Experiment 1, this issue was fixed in Experiment 2. Additionally, analyses indicate that none of the predictors of evaluative shift significantly differed across the comparable low autonomy groups across Experiments 1 and 2 (p 's > 0.269), suggesting the results in Experiment 1 are unlikely to be driven by or greatly affected by this coding error. For more on these analyses, see the Supplement.

midpoint of the 9-point rating scale ($M_{\text{pos}} = 7.27$; $SD_{\text{pos}} = 0.58$; $M_{\text{neg}} = 2.98$; $SD_{\text{neg}} = 0.51$) and standard values for arousal were close to the neutral midpoint ($M_{\text{pos}} = 5.32$; $SD_{\text{pos}} = 0.82$; $M_{\text{neg}} = 5.19$; $SD_{\text{neg}} = 1.05$). The positive and negative picture sets differed in valence, $t(128) = -44.51$, $p < .001$, but not in arousal, $t(110.62) = 0.79$, $p = .430$. The most positive negative US image had a standard value of 3.95; the most negative positive image had a standard value of 6.07.

3.1.3.4. Post-evaluations of CSs. After completing the 60 interactions, participants provided evaluative post-ratings of the 12 CSs, again reacting to the question “What is your impression of this face?” All analyses of evaluative learning are based on the evaluative shift from pre- to post-rating in order to control for pre-existing attitudes. For exploratory purposes, participants also indicated how certain they were regarding their evaluation on a scale from 1 (very uncertain) to 7 (very certain) to serve as an indicator of attitude strength (Krosnick & Petty, 1995; Luttrell & Sawicki, 2020; Tormala & Rucker, 2017).

3.1.3.5. Exploratory post-measures. Next, we assessed a few other exploratory measures. First, participants provided their estimate of the base rate of positive versus negative valence. For that purpose, participants indicated how many of the interactions of each valence they experienced. In accordance with the absolute number of trials, responses in both boxes had to sum up to 60. Second, participants estimated the number of times they sampled each of the 12 CSs. Participants were shown 12 CSs on one screen with a response box next to each of them. Again, responses had to sum up to a total of 60. Third, we assessed participants’ memory for the valence with which each CS was paired. For this purpose, we showed participants the CSs for which they had indicated at least one interaction and asked them to indicate whether they were presented together with positive or negative images. Participants responded in the form of forced-choice drop-down menus. Sampling participants were also asked to list the things that influenced their decisions about which faces to sample using an open-responses format. To conclude the experiment, participants provided demographic information, were debriefed, thanked, and dismissed.

3.1.3.6. Yoked low autonomy condition. For participants in the yoked, low-autonomy condition, the experiment was nearly identical, but with a critical difference. Instead of participants choosing the CSs in the learning task, they were told that the computer would randomly decide with whom they got to interact, when in fact, the sequence was determined by the sampling participant to whom they were yoked. For each participant in the sampling, high-autonomy condition, we saved the sequence of the sampling decisions and the resulting interactions and used those data as the stimulus distribution for the yoked, low-autonomy condition. The identity and sequence of US images was identical to the sampling, high-autonomy participant. The yoked participants’ idiosyncratically selected CSs took on the role of the six positive and six negative CSs of the sampling participants. Participants in the yoked, low-autonomy condition also saw the four-by-three grid alternating with the conditioning trials. However, the screens switched automatically after 2500 ms each.

3.2. Results

Unless stated otherwise, all analyses relied on multi-level modeling to assess relationships within participants on the level of trials. All multi-level models contain random intercepts for participants and CS identities (Judd et al., 2012), as well as fixed effects for the predictors.⁴ Effect sizes are reported in the form of standardized regression coefficients (β)

⁴ CSs that were never sampled/never shown, which was relatively rare (5.21% of all CSs in Experiments 1 and 2), were excluded from analysis.

or relevant means.

3.2.1. Sampling goal manipulation check

We instructed high autonomy participants to sample interactions in a positive, negative, or balanced way. Thus, we first sought to test whether the manipulated instructions impacted sampling behavior over time. We analyzed whether sampling trial (0–59), sampling instructions (Experiment 1: positive = 0.5, negative = −0.5; Experiment 2: positive = 0.5, balanced = 0, negative = −0.5), and their interaction were significant predictors of the sampled valence in a multilevel logistic regression analysis (i.e., with 0 indicating negative US valence and +1 positive US valence).

Experiment 1: As predicted, there was a significant interaction between sampling trial and sampling instructions as a predictor of sampled valence, $b = 0.03$, $SE = 0.004$, $z = 7.58$, $p < .001$. Follow-up analyses probing this interaction show that participants given positive sampling instructions selected more positively-paired CSs over time, $b = 0.02$, $SE = 0.003$, $z = 7.43$, $p < .001$, whereas participants given negative sampling instructions selected more negatively-paired CSs over time, $b = -0.01$, $SE = 0.003$, $z = 3.10$, $p = .002$.

Experiment 2: As predicted and replicating Experiment 1, there was a significant interaction between sampling trial and sampling instructions as a predictor of sampled valence, $b = 0.05$, $SE = 0.004$, $z = 10.76$, $p < .001$. Follow-up analyses probing this interaction show that participants given positive sampling instructions selected more positively-paired CSs over time, $b = 0.03$, $SE = 0.003$, $z = 8.71$, $p < .001$. Participants given balanced sampling instructions did not exhibit a bias to select more positively-paired or negatively-paired CSs over time, $b < 0.001$, $SE = 0.003$, $z = 0.31$, $p = .758$. Participants given negative sampling instructions selected more negatively-paired CSs over time, $b = -0.02$, $SE = 0.003$, $z = 6.48$, $p < .001$. Additionally, the slopes across all three conditions significantly differed from one another (z ’s > 5.11 , p ’s < 0.001).

3.2.2. Confirmatory analysis of evaluative shift

To test our primary hypothesis that the sampling decision effect would be moderated by sampling goal instructions, we predicted evaluative shift as a function of paired valence (positive = 0.5, negative = −0.5), autonomy (high = 0.5, low = −0.5), sampling instructions (Experiment 1: positive = 0.5, negative = −0.5; Experiment 2: positive = 0.5, balanced = 0, negative = −0.5⁵), the number of samples (centered across participants and CSs), and their interactions. The critical test of our hypothesis is whether the sampling decision effect (captured by the interaction between autonomy condition and number of samples) is further moderated by the three-way interaction with sampling instructions.

Experiment 1: As hypothesized, there was a significant interaction between autonomy condition, sampling instructions, and number of samples, $b = 7.19$, $SE = 1.51$, $t(1521.66) = 4.77$, $p < .001$, $\beta = 0.53$, which was not further moderated by paired valence, $b = -3.88$, $SE = 3.19$, $t(1590.59) = 1.22$, $p = .225$, $\beta = -0.28$. The nature of this interaction, which we further probe below, can be viewed in Fig. 1. See

⁵ We used this coding in the primary analysis as our hypothesis is most clearly tested by the comparison between the extreme conditions (i.e., positive vs. negative sampling). Follow-up analyses decomposing significant interactions with the sampling instructions factor converted this single predictor into two dummy-coded predictors that allow for testing the effects of other predictors (e.g., sampling frequency) within a specific sampling instruction condition and testing differences between each instruction condition individually. For example, to test effects in the positive sampling condition, we included both a negative instructions variable (positive = 0, balanced = 0, negative = 1) and a balanced instructions variable (positive = 0, balanced = 1, negative = 0) in order to set the positive instructions condition as the baseline. We used this approach in all applicable analyses across Experiments 2–4.

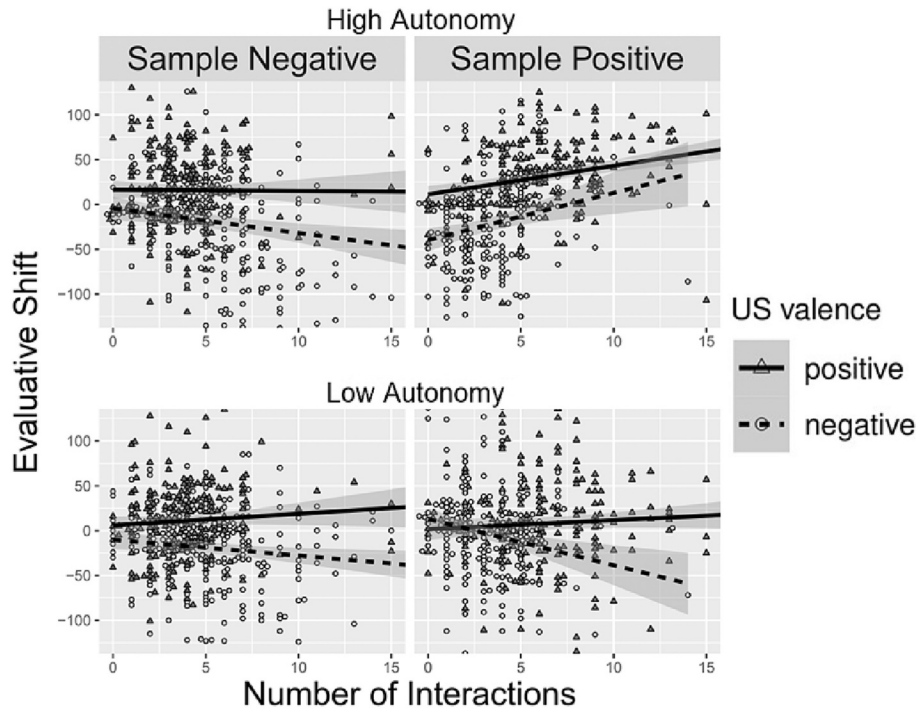


Fig. 1. Evaluative Shift as a Function of Autonomy, Sampling Goal, Paired US Valence, and Number of Samples in Experiment 1.
Note. Values with number of pairings >15 are not shown in the graph to make it more readable. Regression lines are calculated using all values, but without accounting for random effects. Grey areas indicate 95% confidence intervals. Points are offset in dense areas to reduce overplotting.

Table 1
Regression statistics from the confirmatory analysis of evaluative shift in Experiment 1.

	<i>b</i>	<i>SE</i>	<i>t</i> (<i>df</i>)	<i>p</i>
(Intercept)	2.51	2.99	<i>t</i> (155.92) = 0.84	.400
Number of Samples	1.05	0.53	<i>t</i> (1522.10) = 1.96	.050
Paired Valence	38.33	3.51	<i>t</i> (1470.73) = 10.91	<.001
Autonomy Condition	-5.30	4.23	<i>t</i> (156.50) = -1.25	.210
Sampling Instructions	8.46	5.97	<i>t</i> (155.92) = 1.42	.157
N. Samples × Valence	-0.14	1.13	<i>t</i> (1590.58) = -0.12	.901
N. Samples × Autonomy	-1.99	0.76	<i>t</i> (1521.66) = -2.64	.008
N. Samples × S. Instructions	5.84	1.07	<i>t</i> (1522.10) = 5.47	<.001
Valence × Autonomy	-12.11	4.98	<i>t</i> (1470.51) = -2.43	.015
Valence × S. Instructions	8.82	7.02	<i>t</i> (1470.73) = 1.26	.209
Autonomy × S. Instructions	-7.01	8.46	<i>t</i> (156.50) = -0.83	.407
N. Samples × Valence × Autonomy	3.86	1.60	<i>t</i> (1590.59) = 2.42	.016
N. Samples × Valence × S. Instructions	-2.30	2.26	<i>t</i> (1590.58) = -1.02	.307
N. Samples × Autonomy × S. Instructions	-7.20	1.51	<i>t</i> (1521.66) = -4.77	<.001
Valence × Autonomy × S. Instructions	-21.76	9.96	<i>t</i> (1470.51) = -2.19	.029
N. Samples × Valence × Autonomy × S. Instructions	3.88	3.19	<i>t</i> (1590.59) = 1.21	.220

Note. The hierarchical model predicts evaluative shift from number of samples, paired valence, autonomy condition, sampling instructions condition, and their interactions, while including participants and stimulus as random intercepts. The critical interaction testing the primary hypothesis is highlighted in grey for ease of identification.

Table 1 for the regression statistics.
Follow-up analyses decomposing the three-way interaction between autonomy condition, sampling instructions, and number of samples revealed that there was a significant interaction between number of samples and sampling instructions in the high autonomy condition, $b = 5.84$, $SE = 1.07$, $t(1522.10) = 5.47$, $p < .001$, $\beta = 0.43$, but not the low autonomy condition, $b = -1.36$, $SE = 1.07$, $t(1521.23) = 1.27$, $p = .204$, $\beta = -0.10$. Among participants given positive sampling instructions in the high autonomy condition, number of samples predicted a positive evaluative shift, $b = 3.97$, $SE = 0.86$, $t(1531.58) = 4.63$, $p < .001$, $\beta = 0.29$. Among participants given negative sampling instructions in the high autonomy condition, number of samples predicted a negative

evaluative shift, $b = -1.87$, $SE = 0.64$, $t(1503.34) = 2.94$, $p = .003$, $\beta = -0.14$.
Experiment 2: As hypothesized and replicating Experiment 1, there was a significant interaction between autonomy condition, sampling instructions, and number of samples, $b = 5.08$, $SE = 1.04$, $t(2121.51) = 4.86$, $p < .001$, $\beta = 0.55$, which was not further moderated by paired valence, $b = -3.23$, $SE = 2.24$, $t(2114.42) = 1.44$, $p = .150$, $\beta = -0.35$. The nature of this interaction, which we further probe below, can be viewed in Fig. 2. See Table 2 for the regression statistics.
Follow-up analyses decomposing the three-way interaction between autonomy condition, sampling instructions, and number of samples revealed that there was a significant interaction between number of

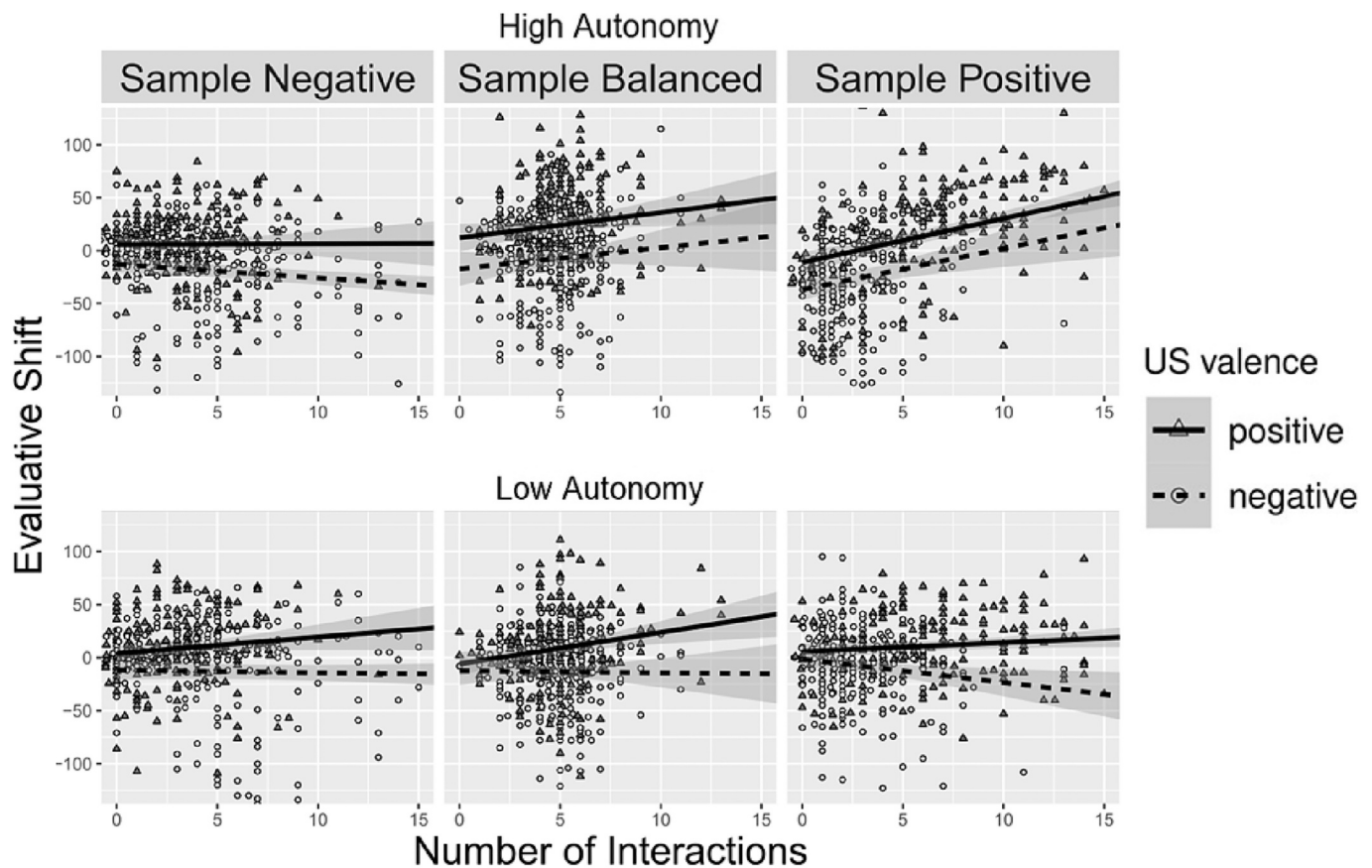


Fig. 2. Evaluative Shift as a Function of Autonomy, Sampling Goal, Paired US Valence, and Number of Samples in Experiment 2.

Note. Values with number of pairings >15 are not shown in the graph to make it more readable. Regression lines are calculated using all values, but without accounting for random effects. Grey areas indicate 95% confidence intervals. Points are offset in dense areas to reduce overplotting.

samples and sampling instructions in the high autonomy condition, $b = 3.56$, $SE = 0.74$, $t(2122.67) = 4.82$, $p < .001$, $\beta = 0.38$.⁶ Among participants given positive sampling instructions in the high autonomy condition, number of samples predicted a positive evaluative shift, $b = 3.40$, $SE = 0.54$, $t(2095.92) = 6.30$, $p < .001$, $\beta = 0.37$. The effect of number of samples significantly differed between participants given positive sampling instructions and those given negative sampling instructions, $b = 3.50$, $SE = 0.80$, $t(2146.77) = 4.38$, $p < .001$, $\beta = 0.38$, such that the positive impact of number of samples observed among participants with a positive sampling goal was eliminated among participants with a negative sampling goal, $b = -0.10$, $SE = 0.59$, $t(2155.11) = 0.18$, $p = .859$, $\beta = -0.01$. Among participants given balanced sampling instructions in the high autonomy condition, there tended to be an overall positive impact of number of samples, though the effect was not significant, $b = 1.61$, $SE = 0.83$, $t(1997.87) = 1.94$, $p = .052$, $\beta = 0.17$.

3.2.3. Valence memory exploratory analysis

We assessed valence memory for all CSs with which participants had at least one interaction. We conducted a logistic regression to predict the accuracy of their memory (1 = correct, 0 = incorrect) from paired

valence (positive = 0.5, negative = -0.5), autonomy (high = 0.5, low = -0.5), sampling instructions (Experiment 1: positive = 0.5, negative = -0.5; Experiment 2: positive = 0.5, balanced = 0, negative = -0.5), and number of samples (centered across participants and CSs) as predictors.⁷

Experiment 1: A higher number of samples was related to higher accuracy, $b = 0.08$, $SE = 0.02$, $z = 3.52$, $p < .001$. The main effect of paired valence fell short of significance, $b = 0.23$, $SE = 0.13$, $z = 1.72$, $p = .085$, with a trend toward higher accuracy for positively-paired CSs. There was a significant interaction between paired valence, sampling instructions, and number of samples, $b = 0.29$, $SE = 0.09$, $z = 3.08$, $p = .002$, which was further qualified by a significant four-way interaction with autonomy condition, $b = 0.75$, $SE = 0.19$, $z = 3.99$, $p < .001$. No other effects were significant, $|z|$'s ≤ 1.54 , p 's ≥ 0.124 .

In the high autonomy condition, there was a significant three-way interaction between sampling instructions, paired valence, and number of samples, $b = 0.67$, $SE = 0.14$, $z = 4.61$, $p < .001$. Among participants given a positive sampling goal, paired valence significantly moderated the effect of number of samples, $b = 0.38$, $SE = 0.11$, $z = 3.41$, $p = .001$, such that a higher number of samples was related to better memory for positively-paired CSs, $b = 0.22$, $SE = 0.07$, $z = 3.11$, $p = .002$, but not negatively-paired CSs, $b = -0.15$, $SE = 0.08$, $z = 1.77$, $p = .077$. Among participants given a negative sampling goal, paired

⁶ There was also an unexpected significant interaction between number of samples and instruction condition in the low autonomy condition, $b = -1.51$, $SE = 0.74$, $t(2177.90) = 2.06$, $p = .040$, $\beta = -0.16$. Though we recommend caution in interpret this effect as it barely reaches the threshold of significance, was not observed in Experiment 1, and was not theoretically-derived or predicted a priori. Nonetheless, analyses probing this interaction can be found in the Supplement.

⁷ Across experiments, predicting valence memory while including random intercepts for participants and CS identities occasionally resulted in over-parameterized models that failed to converge. Thus, although the models with and without these random effects produced very similar results, for all experiments, we report analyses predicting valence memory that do not include random intercepts for participants and CS identities.

Table 2

Regression statistics from the confirmatory analysis of evaluative shift in Experiment 2.

	<i>b</i>	<i>SE</i>	<i>t(df)</i>	<i>p</i>
(Intercept)	-0.58	1.83	<i>t</i> (103.98) = -0.32	.750
Number of Samples	0.86	0.25	<i>t</i> (2139.72) = 3.44	.001
Paired Valence	28.63	1.76	<i>t</i> (2007.77) = 16.24	<.001
Autonomy Condition	1.56	2.96	<i>t</i> (209.85) = 0.53	.598
Sampling Instructions	1.28	3.68	<i>t</i> (220.21) = 0.35	.729
N. Samples × Valence	2.06	0.52	<i>t</i> (2158.65) = 3.96	<.001
N. Samples × Autonomy	1.53	0.5	<i>t</i> (2124.98) = 3.06	.002
N. Samples × S. Instructions	1.02	0.52	<i>t</i> (2119.31) = 1.95	.051
Valence × Autonomy	6.38	3.52	<i>t</i> (2002.30) = 1.81	.070
Valence × S. Instructions	-2.08	4.49	<i>t</i> (2022.77) = -0.46	.643
Autonomy × S. Instructions	8.51	7.37	<i>t</i> (220.78) = 1.15	.248
N. Samples × Valence × Autonomy	-1.79	1.04	<i>t</i> (2167.92) = -1.72	.086
N. Samples × Valence × S. Instructions	0.26	1.12	<i>t</i> (2117.38) = 0.23	.815
N. Samples × Autonomy × S. Instructions	5.08	1.04	<i>t</i> (2121.51) = 4.86	<.001
Valence × Autonomy × S. Instructions	9.27	8.97	<i>t</i> (2014.21) = 1.03	.301
N. Samples × Valence × Autonomy × S. Instructions	-3.23	2.24	<i>t</i> (2114.42) = -1.44	.150

Note. The hierarchical model predicts evaluative shift from number of samples, paired valence, autonomy condition, sampling instructions condition, and their interactions, while including participants and stimulus as random intercepts. The critical interaction testing the primary hypothesis is highlighted in grey for ease of identification.

valence significantly moderated the effect of number of samples in the opposite direction, $b = -0.28$, $SE = 0.09$, $z = -3.10$, $p = .002$, such that a higher number of samples was related to better memory for *negatively*-paired CSs, $b = 0.27$, $SE = 0.07$, $z = 3.86$, $p < .001$, but not positively-paired CSs, $b = -0.01$, $SE = 0.06$, $z = 0.18$, $p = .861$.

In the low autonomy condition, in contrast, number of samples positively predicted accurate valence memory, $b = 0.08$, $SE = 0.03$, $z = 2.71$, $p = .007$, regardless of paired valence, $b = -0.08$, $SE = 0.06$, $z = 1.26$, $p = .206$, sampling instruction condition, $b = -0.01$, $SE = 0.05$, $z = 0.18$, $p = .861$, or the interaction between the three, $b = -0.08$, $SE = 0.12$, $z = -0.69$, $p = .488$. Indeed, number of samples was the only significant predictor of accurate valence memory in the low autonomy condition.

Experiment 2: A higher number of samples was related to higher accuracy, $b = 0.12$, $SE = 0.02$, $z = 6.12$, $p < .001$. There was not a main effect of paired valence, $b = 0.04$, $SE = 0.11$, $z = 0.33$, $p = .740$. There was a significant interaction between paired valence and sampling instructions, $b = -0.64$, $SE = 0.30$, $z = 2.15$, $p = .032$, which was further qualified by a three-way interaction with number of samples, $b = 0.24$, $SE = 0.09$, $z = 2.60$, $p = .009$, which was even further qualified by a significant four-way interaction with autonomy condition, $b = 0.70$, $SE = 0.18$, $z = 3.82$, $p < .001$. No other effects were significant, $|z|$'s ≤ 0.95 , p 's ≥ 0.341 .

In the high autonomy condition, there was a significant three-way interaction between sampling instructions, paired valence, and number of samples, $b = 0.59$, $SE = 0.13$, $z = 4.42$, $p < .001$. Among participants given a positive sampling goal, paired valence moderated the effect of number of samples, $b = 0.32$, $SE = 0.09$, $z = 3.41$, $p < .001$, such that a higher number of samples was related to better memory for positively-paired CSs, $b = 0.31$, $SE = 0.06$, $z = 4.83$, $p < .001$, but worse memory for negatively-paired CSs, $b = -0.69$, $SE = 0.30$, $z = 2.30$, $p = .022$. Among participants given a negative sampling goal, paired valence moderated the effect of number of samples in the opposite direction, $b = -0.28$, $SE = 0.10$, $z = -2.78$, $p = .005$, such that a higher number of samples was related to better memory for *negatively*-paired CSs, $b = 0.24$, $SE = 0.06$, $z = 3.81$, $p < .001$, but not positively-paired CSs, $b = -0.03$, $SE = 0.08$, $z = -0.43$, $p = .665$. Among participants given a balanced sampling goal, there was not a significant interaction between paired valence and number of samples, $b = 0.10$, $SE = 0.12$, $z = -0.80$, $p = .424$. Instead, there was simply a main effect of number of samples such that more samples were related to better memory, $b = 0.13$, $SE = 0.06$, $z = 2.16$, $p = .031$.

In the low autonomy condition, in contrast, number of samples

positively predicted accurate valence memory, $b = 0.11$, $SE = 0.03$, $z = 4.08$, $p < .001$, regardless of paired valence, $b = -0.02$, $SE = 0.06$, $z = -0.27$, $p = .791$, sampling instruction condition, $b = 0.02$, $SE = 0.06$, $z = 0.37$, $p = .711$, or the interaction between the three, $b = -0.11$, $SE = 0.13$, $z = -0.89$, $p = .372$. Indeed, number of samples was the only significant predictor of accurate valence memory in the low autonomy condition.

3.3. Discussion

Experiments 1 and 2 provided support for the hypothesis that the sampling decision effect is dependent upon sampling goal: sampling a stimulus more frequently predicted a more positive evaluative shift among high autonomy participants given a positive sampling goal, but critically, not among high autonomy participants given a negative sampling goal. These findings suggest that sampling a stimulus does not merely shift evaluation via a direct association with positivity. Indeed, even for those CSs which were consistently paired with positive USs, sampling them more frequently did not lead to a positive evaluative shift among high autonomy participants who were given the goal to sample negatively.⁸

4. Experiments 3a and 3b

Experiments 1 and 2 demonstrated the malleability of the sampling decision effect, highlighting the importance of people's sampling goals on their interpretation of their sampling behavior. Next, in a pair of pre-registered experiments, we sought to replicate these effects while also providing insight to the malleability of the impact of choosing *not* to sample a stimulus. We used the same sampling instructions as in Experiment 2 but modified the sampling paradigm so that participants could only select half of the CSs to interact with, which also constitutes choosing half of the CSs *not* to interact with. Previous work using this paradigm found that high autonomy participants demonstrated a negative evaluative shift toward the stimuli they avoided sampling (Hütter et al., 2022). If such an effect is due merely to an association

⁸ That is, number of samples did not significantly predict evaluative shift for positively-paired CSs among high autonomy participants with a negative sampling goal in either Experiment 1, $b = -1.37$, $SE = 1.07$, $t(1575.97) = 1.28$, $p = .200$, $\beta = -0.03$, or Experiment 2, $b = 0.86$, $SE = 1.15$, $t(2145.89) = 0.75$, $p = .455$, $\beta = 0.09$.

between not sampling a stimulus and negativity, we should expect to replicate this effect regardless of instruction condition. However, if as we hypothesize, this effect is subject to peoples' understanding of what (not) sampling a stimulus means, we should find the strongest negative evaluative shift for unsampled CSs when high autonomy participants' goal is to sample positively paired stimuli, but eliminate or reverse this effect when their goal is to sample negatively paired stimuli.

4.1. Method

4.1.1. Participants

Using the same approach as in Experiments 1 and 2, we approximated the required sample size via a power analysis based on a simplified mixed-design ANOVA. Using this approach, we determined 276 participants in Experiment 3a would be required to detect a small effect of size $f = 0.10$ at adequate power ($1 - \beta = 0.80$), and 504 participants in Experiment 3b would be required to detect a small effect of size $f = 0.10$ at high power ($1 - \beta = 0.95$; [Faul et al., 2007](#)). We aimed to recruit slightly above these amounts to cope with potential data loss due to technical or human error.

We received 301 participant responses in Experiment 3a via Prolific (www.prolific.co). We also received 510 participant responses in Experiment 3b on Prolific as a direct replication of Experiment 3a. As the materials were identical across experiments, we integrated the data for presentation and analysis using an Integrative Data Analysis approach ([Curran & Hussong, 2009](#)).⁹ In both experiments, we only accepted participants with an approval rate above 90% who were between 18 and 60 years old and indicated English as their first language. The experiments lasted roughly 15 min and participants were paid 1.60£ for their participation. As the low-autonomy condition could only be run once data from some high-autonomy participants was available, we ran conditions in alternating blocks of participants. Thirty-one responses were excluded due to a technical error in the yoking procedure, resulting in 390 valid participant pairs (364 women, 412 men, 4 preferred to self-identify, $M_{\text{age}} = 37.64$, $SD_{\text{age}} = 11.58$). Thus, based on the final sample size, sensitivity analyses using the same approach we employed to initially determine sample size suggest the smallest effect size we could detect at 80% power ($\alpha = 0.05$) would be $f = 0.064$ ([Faul et al., 2007](#)).

4.1.2. Design

Experiments 3a and 3b implemented a 2 (paired valence: positive vs. negative) $\times$ 2 (autonomy: high vs. low) $\times$ 3 (sampling instructions: positive vs. balanced vs. negative) mixed design with autonomy and sampling instructions as between-participants factors. Additionally, the number of samples per CS was introduced as a continuous quasi-experimental predictor.

4.1.3. Materials and procedure

The overall setup was similar to Experiment 2. However, in these experiments, participants in the high autonomy condition were only allowed to interact with half of the presented CSs. They were still free in their sampling decisions, but once they had interacted with six different CSs, their remaining interactions had to be with those CSs already sampled. The first six unique CSs a participant interacted with were highlighted with a green border, indicating the possibility of future interactions. The remaining unselected CSs were highlighted with a red border and image opacity was decreased, indicating interactions with those CSs were no longer possible. To keep the average number of

samples with paired CSs equal to the first two experiments, the sampling phase was reduced to 30 interactions. Out of the six CSs that participants interacted with, three were paired positively and three were paired negatively.

4.2. Results

Data analysis proceeded as in Experiments 1 and 2, with the exceptions of including experiment as a factor (Experiment 3a = -0.5 , Experiment 3b = 0.5) in all analyses to verify effects were not uniquely driven by one sample or the other, as well as the addition of an analysis for the unsampled CSs.

4.2.1. Sampling goal manipulation check

We first sought to test whether the manipulated instructions impacted sampling behavior over time. We analyzed whether sampling trial (0–59), sampling instructions (positive = 0.5 , balanced = 0 , negative = -0.5), and their interaction were significant predictors of the sampled valence in a multilevel logistic regression analysis (i.e., with 0 indicating negative US valence and +1 positive US valence).

As predicted, there was a significant interaction between sampling trial and sampling instructions as a predictor of sampled valence, $b = 0.07$, $SE = 0.006$, $z = 12.05$, $p < .001$. Follow-up analyses probing this interaction show that participants given positive sampling instructions selected more positively-paired CSs over time, $b = 0.04$, $SE = 0.004$, $z = 10.17$, $p < .001$. Participants given balanced sampling instructions also exhibited a bias to select more positively-paired CSs over time, $b = 0.01$, $SE = 0.004$, $z = 3.65$, $p < .001$, though to a significantly lesser extent than participants given positive sampling instructions, $b = 0.03$, $SE = 0.01$, $z = 4.98$, $p < .001$. Participants given negative sampling instructions selected more negatively-paired CSs over time, $b = -0.03$, $SE = 0.004$, $z = 6.74$, $p < .001$. Additionally, the slopes across all three conditions significantly differed from one another (z 's > 4.98 , p 's < 0.001).

4.2.2. Confirmatory analysis of evaluative shift for paired CSs

To test our primary hypothesis that the sampling decision effect would be moderated by sampling goal instructions, we predicted evaluative shift as a function of paired valence (positive = 0.5 , negative = -0.5), autonomy (high = 0.5 , low = -0.5), sampling instructions (Experiment 1: positive = 0.5 , negative = -0.5 ; Experiment 2: positive = 0.5 , balanced = 0 , negative = -0.5), the number of samples (centered across participants and CSs), and their interactions.

As hypothesized and replicating Experiments 1 and 2, there was a significant interaction between autonomy condition, sampling instructions, and number of samples, $b = 5.36$, $SE = 1.40$, $t(4064.74) = 3.83$, $p < .001$, $\beta = 0.23$, which was not further moderated by paired valence, $b = -3.61$, $SE = 3.22$, $t(4002.75) = 1.12$, $p = .263$, $\beta = -0.15$. The nature of this interaction, which we further probe below, can be viewed in [Fig. 3](#). See [Table 3](#) for the regression statistics.

Follow-up analyses decomposing the three-way interaction between autonomy condition, sampling instructions, and number of samples revealed that there was a significant interaction between number of samples and sampling instructions in the high autonomy condition, $b = 5.26$, $SE = 0.99$, $t(4063.16) = 5.31$, $p < .001$, $\beta = 0.22$, but not in the low autonomy conditions, $b = -0.11$, $SE = 0.99$, $t(4056.74) = -0.11$, $p < .914$, $\beta = -0.004$. Among participants given positive sampling instructions in the high autonomy condition, number of samples predicted a positive evaluative shift, $b = 4.69$, $SE = 0.74$, $t(4269.24) = 6.35$, $p < .001$, $\beta = 0.20$. The effect of number of samples significantly differed between the positive and negative sampling conditions, $b = 6.01$, $SE = 1.04$, $t(4113.71) = 5.76$, $p < .001$, $\beta = 0.26$, such that the positive impact of number of samples was eliminated and tended to reverse in the negative sampling condition, $b = -1.32$, $SE = 0.73$, $t(3938.89) = 1.79$, $p = .073$, $\beta = -0.06$. Among participants given balanced sampling instructions in the high autonomy condition, number of samples did not

⁹ This approach allows the analyses to benefit from the full power of collected samples of data while also including study as a factor to verify that the results are not driven solely by one sample or the other. Indeed, none of the primary results presented significantly differed by study. More information about these analyses is available in the supplement.

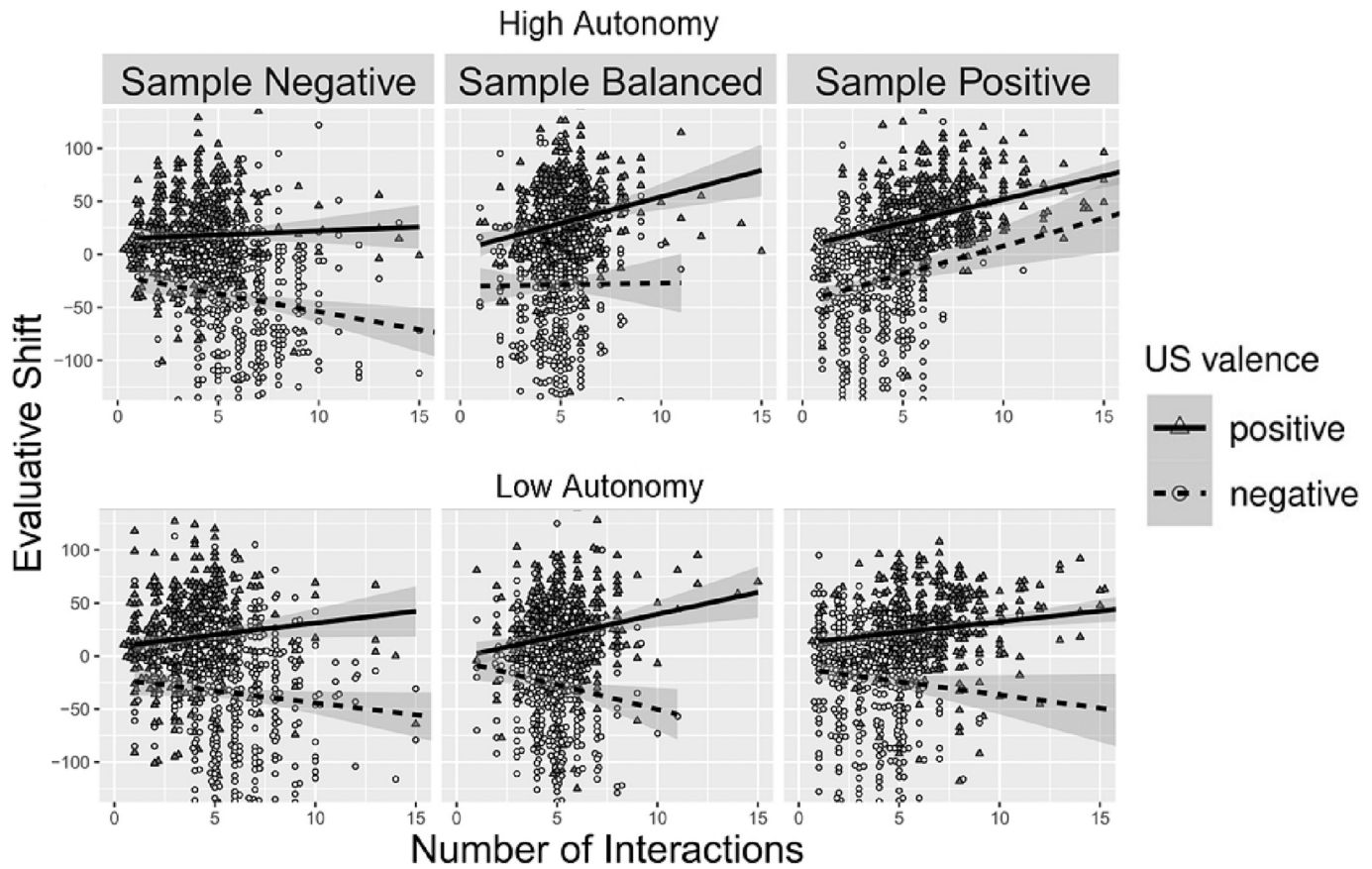


Fig. 3. Evaluative Shift for Paired CSs as a Function of Autonomy, Sampling Goal, Paired US Valence, and Number of Samples in Experiment 3.

Note. Values with number of pairings >15 are not shown in the graph to make it more readable. Regression lines are calculated using all values, but without accounting for random effects. Grey areas indicate 95% confidence intervals. Points are offset in dense areas to reduce overplotting.

significantly predict evaluative shift, $b = 2.00$, $SE = 1.22$, $t(3877.22) = 1.64$, $p = .101$, $\beta = 0.09$.

4.2.3. Confirmatory Analysis of Evaluative Shift for Unpaired CSs

To test our hypothesis that the effect of choosing *not* to sample a stimulus would be moderated by sampling goal instructions, we predicted evaluative shift for unpaired CSs from autonomy condition, sampling instructions, and their interaction. As hypothesized, there was a significant interaction between autonomy condition and sampling instructions, $b = -8.01$, $SE = 3.91$, $t(774.87) = -2.05$, $p = .041$, $\beta = -0.23$. The nature of this interaction, which we further probe below, can be viewed in Fig. 4. See Table 4 for the regression statistics.

In the low autonomy condition, evaluative shift did not differ by sampling instruction condition, $b = 0.27$, $SE = 2.76$, $t(771.73) = 0.10$, $p = .922$, $\beta = 0.01$, nor did it significantly differ from 0 overall, $b = -1.49$, $SE = 1.91$, $t(90.89) = -0.78$, $p = .437$, $\beta = 0.05$. That is, participants in the low autonomy condition showed no evidence of evaluative shift for unpaired CSs, regardless of the sampling instructions that were used to generate the sample.

In the high autonomy condition, in contrast, evaluative shift for unpaired CSs significantly differed depending on sampling instructions, $b = -7.74$, $SE = 2.77$, $t(778.10) = -2.80$, $p = .005$, $\beta = 0.23$. In the positive sampling condition, unpaired CSs became significantly more negative, $b = -7.14$, $SE = 2.51$, $t(239.60) = -2.84$, $p = .005$, $\beta = -0.11$. This significantly differed from the negative sampling condition, $b = -7.72$, $SE = 2.77$, $t(774.01) = -2.79$, $p = .005$, $\beta = 0.22$, in which the effect on evaluative shift was completely eliminated and, if anything, began to reverse, $b = 0.59$, $SE = 2.48$, $t(230.19) = 0.24$, $p = .813$, $\beta = 0.11$. The effect on evaluative shift in the balanced sampling condition, showed a negative non-significant trend, $b = -4.31$, $SE = 2.50$, t

$(237.75) = 1.72$, $p = .086$, $\beta = -0.03$.

4.2.4. Exploratory analysis of valence memory

We assessed valence memory for all CSs with which participants had at least one interaction. We conducted a logistic regression to predict the accuracy of their memory (1 = correct, 0 = incorrect) from paired valence (positive = 0.5, negative = -0.5), autonomy (high = 0.5, low = -0.5), sampling instructions (Experiment 1: positive = 0.5, negative = -0.5; Experiment 2: positive = 0.5, balanced = 0, negative = -0.5), and number of samples (centered across participants and CSs) as predictors.

A higher number of samples was related to higher accuracy, $b = 0.20$, $SE = 0.03$, $z = 7.52$, $p < .001$. There was also a main effect of paired valence such that there was higher accuracy for positively- vs. negatively-paired CSs, $b = 0.36$, $SE = 0.10$, $z = 3.75$, $p < .001$, and a main effect of condition such that high autonomy (vs. low autonomy) participants exhibited greater accuracy, $b = 0.26$, $SE = 0.13$, $z = 1.98$, $p = .048$. However, these effects were qualified by a significant four-way interaction between autonomy condition, sampling instructions, paired valence, and number of samples, $b = 1.01$, $SE = 0.25$, $z = 4.02$, $p < .001$.

In the high autonomy condition, there was a significant three-way interaction between sampling instructions, paired valence, and number of samples, $b = 1.10$, $SE = 0.19$, $z = 5.67$, $p < .001$. Among participants with a positive sampling goal, paired valence moderated the effect of number of samples, $b = 0.65$, $SE = 0.15$, $z = 4.35$, $p < .001$, such that a higher number of samples was related to better memory for positively-paired CSs, $b = 0.61$, $SE = 0.12$, $z = 5.15$, $p < .001$, but not for negatively-paired CSs, $b = -0.01$, $SE = 0.09$, $z = 0.07$, $p = .943$. Among participants with a negative sampling goal, paired valence moderated the effect of number of samples in the opposite direction, $b = -0.46$, $SE = 0.13$, $z = -3.63$, $p < .001$, such that a higher number of samples was

Table 3

Regression statistics from the confirmatory analysis of evaluative shift for paired CSs in Experiment 3.

	<i>b</i>	<i>SE</i>	<i>t(df)</i>	<i>p</i>
(Intercept)	-2.46	1.36	<i>t</i> (98.12) = -1.81	.074
Number of Samples	0.93	0.34	<i>t</i> (4099.70) = 2.74	.006
Paired Valence	52.44	1.44	<i>t</i> (3892.02) = 36.34	< .001
Autonomy Condition	2.39	2.05	<i>t</i> (886.40) = 1.16	.244
Sampling Instructions	7.85	2.55	<i>t</i> (928.35) = 3.08	.002
N. Samples * Valence	3.37	0.76	<i>t</i> (4328.25) = 4.44	< .001
N. Samples * Autonomy	1.87	0.68	<i>t</i> (4099.23) = 2.77	.006
N. Samples * S. Instructions	2.58	0.70	<i>t</i> (4055.24) = 3.68	< .001
Valence * Autonomy	7.08	2.89	<i>t</i> (3902.38) = 2.45	.014
Valence * S. Instructions	-5.56	3.62	<i>t</i> (3911.81) = -1.54	.124
Autonomy * S. Instructions	6.70	5.09	<i>t</i> (929.01) = 1.37	.170
N. Samples * Valence * Autonomy	-2.92	1.52	<i>t</i> (4328.87) = -1.92	.055
N. Samples * Valence * S. Instructions	-2.77	1.61	<i>t</i> (3999.25) = -1.72	.086
N. Samples * Autonomy * S. Instructions	5.36	1.40	<i>t</i> (4064.74) = 3.83	< .001
Valence * Autonomy * S. Instructions	1.26	7.23	<i>t</i> (3915.05) = 0.18	.861
N. Samples * Valence * Autonomy * S. Instructions	-3.61	3.22	<i>t</i> (4002.75) = -1.12	.263

Note. The hierarchical model predicts evaluative shift from number of samples, paired valence, autonomy condition, sampling instructions condition, and their interactions, while including participants and stimulus as random intercepts. The full model also includes Experiment (3a vs. 3b) and its interactions with the other predictors. For ease of presentation, we have omitted the regression statistics for this factor here, but have included them in an additional table in the Supplement. The critical interaction testing the primary hypothesis is highlighted in grey for ease of identification.

related to better memory for negatively-paired CSs, $b = 0.40$, $SE = 0.09$, $z = 4.52$, $p < .001$, but not positively-paired CSs, $b = -0.03$, $SE = 0.08$, $z = -0.36$, $p = .723$. Among participants with a balanced sampling goal, there was not a significant interaction between paired valence and number of samples, $b = -0.06$, $SE = 0.19$, $z = -0.33$, $p = .739$. Instead, there was simply a main effect of number of samples such that more samples were related to better memory, $b = 0.67$, $SE = 0.23$, $z = 2.91$, $p = .004$.

In the low autonomy condition, in contrast, number of samples positively predicted accurate valence memory, $b = 0.16$, $SE = 0.03$, $z = 5.31$, $p < .001$. However, this effect was moderated by paired valence, $b = -0.02$, $SE = 0.06$, $z = -0.27$, $p = .791$, such that number of samples had a larger impact for negatively-paired CSs, $b = 0.26$, $SE = 0.05$, $z = 4.89$, $p < .001$, than for positively-paired CSs, $b = 0.10$, $SE = 0.05$, $z = 2.15$, $p = .031$.

4.3. Discussion

Experiments 3a and 3b replicated the findings from Experiments 1 and 2: high autonomy participants showed a positive evaluative shift for CSs when they had the goal to sample for positivity, but not when they had the goal to sample for negativity. Additionally, these experiments extend the previous studies by showing complementary effects among CSs participants chose not to sample, with high autonomy participants showing a negative evaluative shift for unsampled CSs when they had the goal to sample for positivity, but not for negativity. Thus, these results further support the notion that the sampling decision effect is sensitive to people's interpretation of what their sampling behavior implies based on their sampling goals.

5. Experiment 4

In the previous experiments, we directly manipulated participants sampling goals by explicitly asking them to primarily sample for positivity, negativity, or in a balanced way. Though this manipulation is valuable from a theoretical stance in providing a clear manipulation of sampling behavior, one may wonder about the applicability of the results outside the experimental setting, especially given the seemingly artificial goal to sample for negativity. Yet, we would note that while the

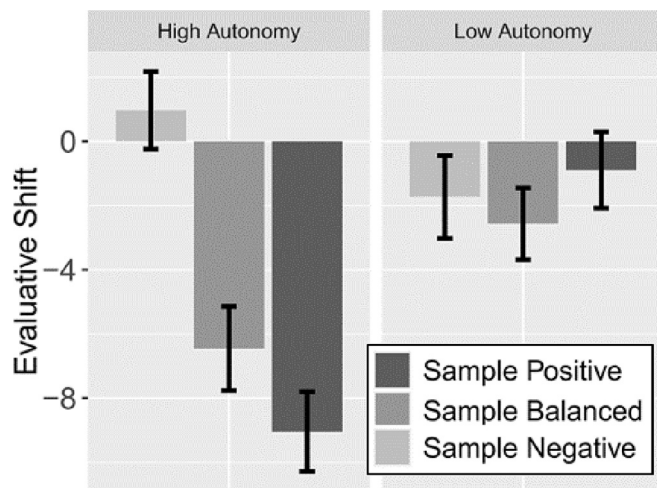


Fig. 4. Average Evaluative Shift as a Function of Autonomy and Sampling Goal for Stimuli that Were Never Paired in Experiment 3.

Note. Error bars indicate standard errors.

Table 4

Regression statistics from the confirmatory analysis of evaluative shift for unpaired CSs in Experiment 3.

	<i>b</i>	<i>SE</i>	<i>t(df)</i>	<i>p</i>
Intercept	-2.56	1.74	<i>t</i> (62.81) = 1.47	.146
Autonomy Condition	-2.12	1.60	<i>t</i> (772.45) = 1.33	.184
Sampling Instructions	-3.73	1.95	<i>t</i> (774.96) = 1.91	.056
Autonomy * S. Instructions	-8.01	3.91	<i>t</i> (774.87) = 2.05	.041

Note. The hierarchical model predicts evaluative shift for unpaired CSs from autonomy condition, sampling instructions condition, and their interaction, while including participants and stimulus as random intercepts. The full model also includes Experiment (3a vs. 3b) and its interactions with the other predictors. For ease of presentation, we have omitted the regression statistics for this factor here, but have included them in an additional table in the Supplement. The critical interaction testing the primary hypothesis is highlighted in grey for ease of identification.

default in everyday life may be to sample for positivity, there are indeed situations in which people hold goals to sample for negativity: police officers seek out crime to stop it, firefighters approach dangerous fires to put them out, doctors look for illness and disease to treat them, etc. Building from this insight, a pre-registered Experiment 4 sought to replicate the previous findings demonstrating the malleability of the sampling decision effect using a more naturalistic manipulation of sampling goals that would lead people to sample for positivity, negativity, or in a balanced way. Experiment 4 focused solely on the high autonomy conditions and reframed the task to be an environmentalism game in which participants were tasked with finding examples of companies engaging in environmental protection so participants could encourage them (sample positive), companies engaging in environmental harm so participants could discourage them (sample negative), or a mix of both (sample balanced). We expected to replicate the findings from the previous studies whereby more frequently sampled stimuli would show a more positive evaluative shift when participants had a goal that implied positive sampling, but not when they had a goal that implied negative sampling.

5.1. Method

5.1.1. Participants

Using the same approach as in the previous experiments, we approximated the required sample size via a power analysis based on a mixed-design ANOVA. Using this simplified approach, 264 participants in Experiment 4 would provide high power ($1 - \beta = 0.95$) to detect a small effect of size $f = 0.10$ (Faul et al., 2007). We aimed to recruit slightly above these amounts to cope with potential data loss due to technical or human error.

We recruited using Prolific (www.prolific.co), only accepting participants with an approval rate above 90% who were between 18 and 60 years old and indicated English as their first language. The experiments lasted roughly 15 min and participants were paid 1.80£ for their participation. The site returned data from 268 participants (112 women, 153 men, 3 preferred not to say, $M_{\text{age}} = 38.35$, $SD_{\text{age}} = 11.55$). Thus, based on the final sample size, sensitivity analyses using the same approach we employed to initially determine sample size suggest the smallest effect size we could detect at 80% power ($\alpha = 0.05$) would be $f = 0.087$ (Faul et al., 2007).

5.1.2. Design

Experiment 4 implemented a 2 (paired valence: positive vs. negative) $\times$ 3 (sampling instructions: positive vs. balanced vs. negative) mixed design with sampling instructions as a between-participants factor. Additionally, the number of samples per CS was introduced as a continuous quasi-experimental predictor.

5.1.3. Materials and procedure

The overall setup was similar to Experiment 2 in that high autonomy participants were randomly assigned the goal to sample for positivity, negativity, or in a balanced way, and that the task provided participants autonomy to distribute 60 interactions in any way they wanted over the twelve CSs. However, Experiment 4 included a few modifications. Most importantly, Experiment 4 changed the framing of the task, as well as the framing of participants' goals in the task, to be more realistic. Specifically, the task was framed as an environmentalism game in which participants would play the role of an environmental regulator, and the 12 CSs were presented as faces of people running different companies. As in the previous experiments, participants selected one of the faces from a grid on each trial to interact with. The CS was then paired with a US, which was either a positive image of environmental protection or a negative image of environmental harm. These US images were pretested on Prolific with $N = 111$ participants (65 women, 46 men, 1 diverse; 18–54 years old, $M = 31.64$, $SD = 9.61$). Participants rated 136 landscapes (and 136 animals) according to pleasure, arousal, and potency on

three 9-point scales. The scales were modeled after the Self-Assessment Manikin scale (Bradley & Lang, 1994) with the exception that the scale allowed intermediate ratings. Each participant rated only a random set of 68 of these stimuli. We selected one set of 30 pleasant stimuli ($M_{\text{pos}} = 7.70$, $SD_{\text{pos}} = 0.31$) and one set of 30 unpleasant stimuli ($M_{\text{neg}} = 3.08$, $SD_{\text{neg}} = 1.38$). The pleasure ratings differed significantly between the groups, $t(58) = 17.55$, $p < .001$. While potency also differed between the two sets ($M_{\text{pos}} = 5.56$, $SD_{\text{pos}} = 0.43$, $M_{\text{neg}} = 4.19$, $SD_{\text{neg}} = 0.33$, $t(58) = 13.49$, $p < .001$), arousal was roughly equal ($M_{\text{pos}} = 4.65$, $SD_{\text{pos}} = 0.33$, $M_{\text{neg}} = 4.45$, $SD_{\text{neg}} = 0.45$, $t(58) = 1.92$, $p = .060$).

Participants were told that the environmental images were effects of each CS's company policies. We manipulated participants' goals by telling some participants (sample positive) that their task as a regulator was to "document reasons to provide tax benefits encouraging companies that are engaging in positive environmental policies. That is, your goal is to encourage as many environmentally helpful actions as possible." Other participants (sample negative) were assigned the goal to "document reasons to apply fines that will discourage companies that are engaging in negative environmental policies. That is, your goal is to discourage as many environmentally harmful actions as possible." Finally, other participants (sample balanced) were given both instructions, highlighting their goal was "to strike a balance between encouraging environmentally helpful actions and discouraging environmentally harmful actions." Because the primary aim of Experiment 4 was to test if our previous effects in the high autonomy conditions replicate with a more realistic task framing and less direct sampling instructions, we only included high autonomy conditions in this experiment.

5.2. Results

Data analysis proceeded as in Experiments 1 and 2, with the exception of no longer including autonomy condition as a factor.

5.2.1. Sampling goal manipulation check

We were interested in whether participants showed a preference for positively or negatively paired CSs depending on the less direct sampling instructions provided in Experiment 4, and how this preference develops over time. We therefore analyzed whether sampling trial (0–59), sampling instructions (positive = 0.5, balanced = 0, negative = -0.5), and their interaction were significant predictors of the sampled valence in a multilevel logistic regression analysis (i.e., with 0 indicating negative US valence and +1 positive US valence).

As predicted, there was a significant interaction between sampling trial and sampling instructions as a predictor of sampled valence, $b = 0.05$, $SE = 0.003$, $z = 17.80$, $p < .001$. Follow-up analyses probing this interaction show that participants given positive sampling instructions selected more positively-paired CSs over time, $b = 0.02$, $SE = 0.002$, $z = 12.23$, $p < .001$. Participants given balanced sampling instructions did not show a significant bias in sampling, $b = 0.003$, $SE = 0.002$, $z = 1.82$, $p = .068$. Participants given negative sampling instructions selected more negatively-paired CSs over time, $b = -0.02$, $SE = 0.002$, $z = 13.00$, $p < .001$. Additionally, the slopes across all three conditions significantly differed from one another (z 's > 7.97 , p 's < 0.001).

5.2.2. Confirmatory analysis of evaluative shift

To test our primary hypothesis that the sampling decision effect would be moderated by sampling goal instructions, we predicted evaluative shift as a function of paired valence (positive = 0.5, negative = -0.5), sampling instructions (positive = 0.5, balanced = 0, negative = -0.5), the number of samples (centered across participants and CSs), and their interactions.

As hypothesized and replicating the previous studies, there was a significant interaction between sampling instructions and number of samples, $b = 2.64$, $SE = 0.62$, $t(3051.95) = 4.28$, $p < .001$, $\beta = 0.17$. However, in Experiment 4, this effect was further moderated by paired

valence, $b = -3.67$, $SE = 1.38$, $t(2997.81) = 2.66$, $p = .008$, $\beta = -0.24$. The nature of this interaction, which we further probe below, can be viewed in Fig. 5. See Table 5 for the regression statistics.

Among participants given a positive sampling goal, number of samples predicted a positive evaluative shift, $b = 3.13$, $SE = 0.46$, $t(3085.83) = 6.77$, $p < .001$, $\beta = 0.21$, regardless of paired valence, $b = -1.86$, $SE = 1.00$, $t(3070.01) = 1.85$, $p = .065$, $\beta = -0.12$. Among participants given a negative sampling goal, this effect was not simply eliminated as in the majority of the previous experiments, but significantly reversed such that number of samples predicted a negative evaluative shift, $b = -1.61$, $SE = 0.54$, $t(3116.49) = 3.01$, $p = .003$, $\beta = -0.11$, regardless of paired valence, $b = 1.26$, $SE = 1.14$, $t(3103.28) = 1.10$, $p = .269$, $\beta = 0.08$.

Finally, the effect of number of samples depended upon paired valence among participants given a balanced sampling goal, $b = 5.62$, $SE = 1.16$, $t(3037.52) = 4.85$, $p < .001$, $\beta = 0.37$. Positively paired CSs became more positive with more samples, $b = 3.09$, $SE = 0.76$, $t(3013.54) = 4.05$, $p < .001$, $\beta = 0.20$, and negatively paired CSs became more negative with more samples, $b = -2.53$, $SE = 0.85$, $t(3018.41) = 2.97$, $p = .003$, $\beta = 0.17$.

5.2.3. Exploratory analysis of valence memory

We assessed valence memory for all CSs with which participants had at least one interaction. We conducted a logistic regression to predict the accuracy of their memory (1 = correct, 0 = incorrect) from paired valence (positive = 0.5, negative = -0.5), sampling instructions (positive = 0.5, balanced = 0, negative = -0.5), and number of samples (centered across participants and CSs) as predictors.

A higher number of samples was related to higher accuracy, $b = 0.14$, $SE = 0.02$, $z = 6.02$, $p < .001$, and positive pairings were more accurately recalled than negative pairings, $b = 0.69$, $SE = 0.11$, $z = 6.48$, $p < .001$. There was also a significant interaction between number of samples and paired valence, $b = 0.11$, $SE = 0.05$, $z = 2.27$, $p = .023$, which was further qualified by a significant three-way interaction with sampling goal, $b = 1.20$, $SE = 0.11$, $z = 10.70$, $p < .001$.

No other effects were significant, $|z|$'s ≤ 1.92 , p 's ≥ 0.055 .

Among participants given a positive sampling goal, paired valence moderated the effect of number of samples, $b = 0.68$, $SE = 0.09$, $z = 8.00$, $p < .001$, such that a higher number of samples was related to better memory for positively-paired CSs, $b = 0.48$, $SE = 0.07$, $z = 7.02$, $p < .001$, but worse memory for negatively-paired CSs, $b = -0.20$, $SE = 0.05$, $z = 4.13$, $p < .001$. Among participants given a negative sampling goal, paired valence moderated the effect of number of samples in the opposite direction, $b = -0.53$, $SE = 0.08$, $z = -6.93$, $p < .001$, such that a higher number of samples was related to better memory for negatively-paired CSs, $b = 0.36$, $SE = 0.04$, $z = 8.16$, $p < .001$, and worse memory for positively-paired CSs, $b = -0.17$, $SE = 0.06$, $z = -2.80$, $p = .005$. Among participants given a balanced sampling goal, there was also a significant interaction between paired valence and number of samples, $b = 0.24$, $SE = 0.09$, $z = 2.64$, $p = .008$. A higher number of samples was related to better memory for positively-paired CSs, $b = 0.32$, $SE = 0.07$, $z = 4.44$, $p < .001$, but not for negatively-paired CSs, $b = 0.09$, $SE = 0.05$, $z = 1.60$, $p = .109$.

5.3. Discussion

Overall, Experiment 4 replicated the findings of the previous studies, further demonstrating the malleability of the sampling decision effect. Critically, Experiment 4 did so by creating a more realistic version of the task that provided participants with different goals that would indirectly lead them to sample for positivity, negativity, or in a balanced way. While one might expect that using this more indirect manipulation would produce weaker results than previous manipulations that directly gave participants the explicit goal to sample for one valence or the other, if anything, the results appear even stronger in Experiment 4 than the previous experiments. Indeed, using a more naturalistic manipulation,

we found that the evaluative impact of sampling decisions is not merely eliminated with a negative sampling goal, but significantly reverses. That is, when participants had the goal to seek out and prevent environmental harm, they showed a negative evaluative shift the more frequently they approached a CS, which was not moderated by the valence of whether the CS was paired with negative environmental harm or positive environmental protection.¹⁰ Interestingly, the pattern of results for the balanced condition also differed in the current experiment from the prior experiments. In the previous experiments, the balanced condition tended to show similar but weaker results to the positive sampling condition. In Experiment 4, the effect of number of samples in the balanced condition was instead dependent upon paired valence, whereby the effect of paired valence was stronger with increasing samples. One possibility for this difference across experiments is the fact that, unlike the previous experiments, Experiment 4 does not represent a mere evaluative conditioning paradigm in which CSs simply co-occur with valent USs. Instead, the task framing causally links the CSs to their paired USs (i.e., the CSs' companies are explicitly labelled as the cause of the environmental protection or harm depicted in the US images).

6. General discussion

The current results suggest that the sampling decision effect cannot merely be explained by an association between sampling and positivity, but instead, is impacted by people's understanding of what their sampling behavior implies based on their sampling goal. Across multiple experiments, participants were given the autonomy to select stimuli (CSs) to interact with that were then paired with positive or negative stimuli (USs). Additionally, participants were given the goal to sample for positivity, negativity, or in a balanced way. Contrary to the possibility that more frequent sampling always leads to more positive evaluations, the current experiments consistently show that this effect is moderated by the provided sampling goals.

More frequent sampling predicted a positive evaluative shift, regardless of paired valence, when people were given the goal to sample for positivity (and, to a weaker extent, typically when they were given the goal to sample in a balanced way). However, when people were given the goal to sample for negativity, this effect was eliminated. Further, when the task was framed in a way that the "sample for negativity" rule was more ecologically valid and relatable, the typical effect reversed such that more frequent sampling predicted a negative evaluative shift, regardless of paired valence. Similarly, while unsampled stimuli showed a negative evaluative shift when people were given the goal to sample positively-paired stimuli (or, again to a weaker extent, to sample in a balanced way), this effect was eliminated when people were given the goal to sample negatively-paired stimuli.

Finally, data from the yoked participants demonstrates that the sampling decision itself is critical to these effects, and that they cannot simply be explained by differences in the different stimuli distributions created via different sampling goals. Indeed, our studies consistently found that the sampling goals that were used to generate the distributions had little impact on low autonomy participants' evaluations of the stimuli. Thus, the current results suggest an important role for people's interpretation of their sampling decisions on their subsequent evaluations.

¹⁰ Though, one should be a bit cautious in interpreting this finding in that, while the effect was not moderated by paired valence, it was also the case that the simple effect of number of samples for positively-paired CSs in this condition trended negative but was not significant, $b = -0.98$, $SE = 1.04$, $t(3131.14) = 0.95$, $p = .342$, $\beta = 0.06$.

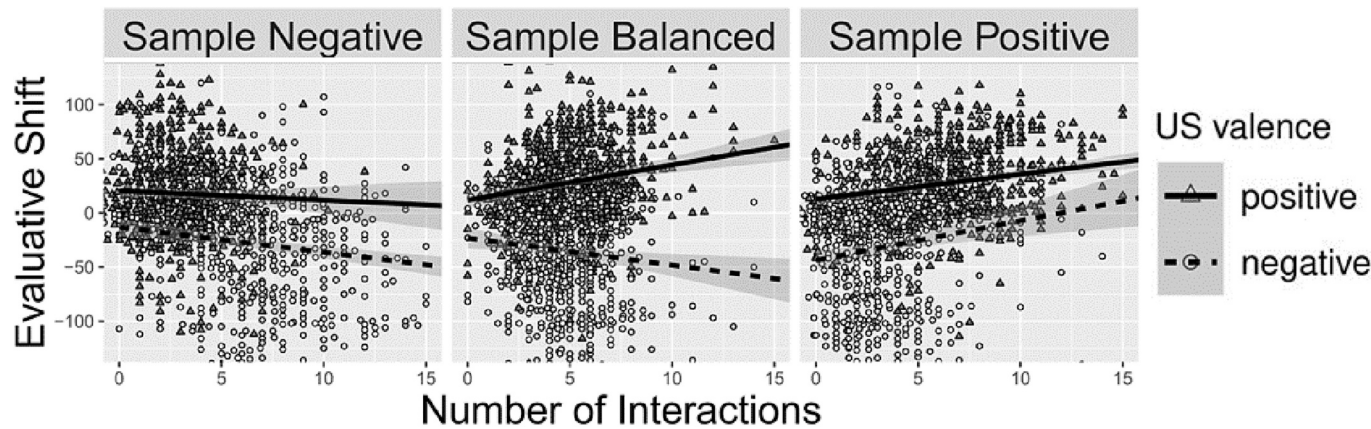


Fig. 5. Evaluative Shift as a Function of Sampling Goal, Paired US Valence, and Number of Samples in Experiment 4.
Note. Values with number of pairings >15 are not shown in the graph to make it more readable. Regression lines are calculated using all values, but without accounting for random effects. Grey areas indicate 95% confidence intervals. Points are offset in dense areas to reduce overplotting.

Table 5
Regression statistics from the confirmatory analysis of evaluative shift in Experiment 4.

	<i>b</i>	<i>SE</i>	<i>t</i> (<i>df</i>)	<i>p</i>
(Intercept)	-2.51	1.61	<i>t</i> (177.88) = -1.56	.120
Number of Samples	0.89	0.3	<i>t</i> (3115.62) = 3.01	.003
Paired Valence	54.73	1.79	<i>t</i> (2910.95) = 30.57	< .001
Sampling Instructions	-0.02	3.93	<i>t</i> (312.19) = -0.01	.995
N. Samples × S. Valence	2.03	0.62	<i>t</i> (3144.63) = 3.26	.001
N. Samples × S. Instructions	2.64	0.62	<i>t</i> (3051.95) = 4.28	< .001
Valence × S. Instructions	8.02	4.56	<i>t</i> (2952.72) = 1.76	.078
N. Samples × Valence × S. Instructions	-3.67	1.38	<i>t</i> (2997.81) = -2.66	.008

Note. The hierarchical model predicts evaluative shift from number of samples, paired valence, sampling instructions condition, and their interactions, while including participants and stimulus as random intercepts. The critical interaction testing the primary hypothesis is highlighted in grey for ease of identification.

6.1. Implications for the Sampling Decision Effect

Previous work using the paradigm employed in the current experiments demonstrated a sampling decision effect whereby more frequently sampled CSs showed a more positive evaluative shift, regardless of whether they were consistently paired with positive or negative USs (Hütter et al., 2022). Such an effect is consistent with prior research arguing that behavior serves as a source of information about one’s attitudes (e.g., Bem, 1972; Cacioppo et al., 1993; De Houwer, Crombez, Baeyens, & Hermans, 2001; Festinger, 1957; Hütter & Genschow, 2020; Van Dessel et al., 2015, 2018). Across experiments, the current work replicates this finding. That is, when people had the goal to sample positively-paired stimuli, more frequently sampling a stimulus predicted a more positive evaluative shift.

However, the current studies extend our understanding of this phenomenon by highlighting the fact that more frequent sampling does not always predict a positive shift in evaluation. In particular, when people were given the goal to sample negatively-paired stimuli, the effect was eliminated or even reversed (as was the case in Experiments 1 and 4) such that more frequent sampling predicted a negative evaluative shift. Such findings suggest that sampling does not necessarily lead to liking in all cases, but rather, that the effect is sensitive to people’s interpretations of what their sampling behavior implies.

This finding is also in line with work in the realm of dissonance theory showing that the effects of people’s past behavior on their subsequent evaluations are sensitive to whether people interpreted the behavior as indicative of their attitudes (e.g., versus viewing a behavior as driven by an external force and thus not informative about one’s

attitudes; Cooper, 1971; Sakai & Andow, 1980). However, the current work highlights how, even when the behavior is viewed as informative, it can predict opposite effects on subsequent evaluations depending on people’s understanding of what sampling behavior is indicative of in a given situation. For instance, most strikingly in Experiment 4, it is not simply the case that sampling behavior was informative in one condition (positive sampling) and not the other (negative sampling), but rather, that it was informative in both conditions and in opposite directions.

Additionally, while it is undoubtedly the case that people typically operate under a frame in which they seek to sample positively-paired stimuli, sampling negatively-paired stimuli is not an entirely ecologically-invalid goal that is only observable in artificial laboratory experiments. That is, there are likely times when people have goals that lead them to engage in active sampling for negativity, as exemplified in Experiment 4. Indeed, it was precisely when the task entailed a more ecologically-valid goal leading people to sample negatively paired stimuli (seek out companies engaging in environmental harm to fine them) that we observed the strongest effects of the negativity sampling goal, leading to a significant reversal whereby increased sampling behavior predicted a negative evaluative shift.

In this way, the current work suggests value in future research exploring the role of sampling decision effects in settings where people’s goals suggest that sampling predicts negativity. For instance, might the fact that a police officer’s job is to sample for negativity (crime) predict negative evaluative shifts for individuals or groups that they sample more frequently (e.g., profiling) even when those individuals/groups end up not to be engaging in a crime? Accordingly, further exploring such situations and the ways in which people’s understanding of what

their sampling behavior implies for evaluation in different situations has the potential to produce findings with important real-world implications.

6.2. Open questions and future directions

6.2.1. The role of valence memory. In the current experiments, we measured participants' valence memory for whether stimuli were paired positively or negatively and found a general pattern in which accuracy depended upon sampling frequency, paired valence, and participants' goals. Among participants given the goal to sample positively-paired stimuli, more frequent sampling was related to better memory for positively-paired CSs and worse memory for negatively-paired CSs. Among participants given the goal to sample negatively-paired stimuli, more frequent sampling was related to worse memory for positively-paired CSs and better memory for negatively-paired CSs. Finally, among participants given the goal to sample in a balanced way, sampling frequency tended to predict better memory overall.

These data can be interpreted in several, but compatible ways. On the one hand, participants likely sample those stimuli more often they remember to serve their sampling goal (e.g., positively-paired stimuli in the positive sampling goal condition). On the other hand, participants likely remember better the paired valence of those stimuli that they sampled more frequently. Finally, knowing that their assigned goal was to sample for positivity (negativity), participants may have simply used this information to infer valence, "Given my goal, if I sampled this face often, it was probably positive (negative)." This third mechanism would contribute to higher accuracy for the paired valence compatible with participants' sampling goal and lower accuracy for the paired valence incompatible with the sampling goal. It seems plausible that each of these mechanisms contributes to our findings, suggesting value in future research directly testing these suggested relationships.

Moreover, it is not clear whether this pattern in memory performance is crucial in the genesis of the sampling decision effect. Stronger memory was accompanied by sampling decision effects on evaluations. Therefore, the question arises whether these effects merely co-occur or are mutually dependent. One possible explanation for a causal relationship between valence memory and the sampling decision effect is that the sampling decision effect is mediated by more accurate memory for the frequently seen stimuli (matching participants' sampling goal) and less accurate memory for the infrequently seen stimuli (mismatching participants' sampling goal). Specifically, participants with the sample for positivity goal might misremember the more (vs. less) frequently sampled negatively-paired CSs as positively-paired and thus like them more, and participants with the sample for negativity goal might misremember the more (vs. less) frequently sampled positively-paired CSs as negatively-paired and thus like them less. Alternatively, the effects on memory and evaluations may occur independently and concurrently. As such, the current findings suggest additional avenues for future research to test these different possibilities against each other.

6.2.2. Evaluative vs. operant conditioning. The current experiments also offer the potential to explore learning effects in a context that pits evaluative and operant conditioning effects against one another. In evaluative conditioning, attitudes (and their corresponding behaviors) toward a stimulus are altered via its association with positivity or negativity. In operant conditioning, behaviors (and their corresponding attitudes) are altered via their potential of producing outcomes consistent ("rewards") or inconsistent ("punishments") with an organism's goals. In particular, as in typical evaluative conditioning experiments, CSs are paired with valent USs, which may themselves shift subsequent behavior, whereby people become more likely to select CSs paired with positive USs and less likely to select CSs paired with negative USs. Indeed, our experiments show evidence for such an effect: across studies, the tendency for people to sample more in line with their assigned goal

over time is stronger for people given a goal to sample positively paired stimuli than those given a goal to sample negatively paired stimuli, $B = 0.01$, $SE = 0.002$, $z = 2.82$, $p = .005$, and across studies, participants given a goal to sample in a balanced way nevertheless tend to show a slight bias toward positivity over time, $B = 0.003$, $SE = 0.001$, $z = 2.36$, $p = .018$. Additionally, regardless of goal and sampling frequency, across experiments, there is a strong and consistent effect of paired valence on evaluative shift such that positively paired CSs are evaluated more positively after the learning procedure than negatively paired CSs.

However, orthogonal to this, the incorporation of sampling instructions also allows us to observe effects of operant conditioning. In particular, participants are given the goal to sample for positivity, negativity, or in a balanced way, and as such, outcomes that match the goal technically act as a reward (i.e., they increase the likelihood of the behavior being performed again) and outcomes that mismatch the goal technically act as a punishment (i.e., they decrease the likelihood of the behavior being performed again; Thorndike, 1927). Indeed, across all studies, we found clear patterns in sampling behavior such that instructions impact sampling behavior over time in this way. In the positive sampling condition, this operant conditioning principle overlaps with evaluative conditioning. However, within the sample negatively condition, this creates a curious situation in which "rewards" constitute negative images (and "punishments" constitute positive images). In other words, operant conditioning and evaluative conditioning represent opposing forces that may influence sampling behavior and evaluations differently. Indeed, CSs associated with "rewards" as determined by the sampling goal are sampled more frequently and CSs associated with "punishments" as determined by the sampling goal are sampled less frequently. Nonetheless, among people given a negative sampling goal, the more (vs. less) frequently "rewarded" CSs are not liked more (i.e., sampling a negative stimulus frequently vs. infrequently does not predict greater liking), but appear to be liked even less. In this way, the current paradigm creates a situation in which frequently rewarding a behavior (i.e., the outcome of sampling a CS is a negative pairing, which makes people more likely to sample the CS again) leads to people evaluating the rewarded stimulus more negatively. Thus, evaluative ratings appear to generally follow principles of evaluative rather than operant conditioning, whereas sampling behavior appears to follow principles of operant rather than evaluative conditioning. In this way, the current research serves as an initial step highlighting the value of future research on rewards and punishments in such combined valence situations as the one implemented in the present paradigm.

6.2.3. Approach-avoidance effects. The current experiments focused on the effect of people's decisions to sample (or not sample) a stimulus on evaluations. However, to the extent that sampling a real-world stimulus often involves approaching it (e.g., sampling food or interactions with an individual), the current findings have the potential to connect with the broader approach-avoidance literature as well. Indeed, mirroring the typical findings utilizing the current paradigm showing that sampling a stimulus tends to have a positive impact on evaluation and not sampling a stimulus tends to have a negative impact (Hütter et al., 2022), a host of studies have demonstrated that merely approaching a stimulus or group causes people to like it more, and merely avoiding a stimulus or group leads people to like it less (Cacioppo et al., 1993; De Houwer et al., 2001; Hütter & Genschow, 2020; Van Dessel et al., 2015, 2018; Van Dessel, De Houwer, Gast, Roets, & Smith, 2020).

Traditionally, these effects were explained by approach-positive and avoid-negative associations attaching valence to approached or avoided stimuli (Cacioppo et al., 1993). Subsequent work questioned the extent to which basic associative processes underlie these effects by highlighting the role of people's interpretation in this process (e.g., Corneille & Stahl, 2019; Van Dessel et al., 2019). For instance, studies have demonstrated approach-avoidance effects could be eliminated if an action was not viewed as approach/avoidance behavior, or that the same

action could have opposite effects depending on if it was framed as approach or avoidance (De Houwer et al., 2001; Hütter & Genschow, 2020; Markman & Brendl, 2005). In this way, the current work further mirrors approach-avoidance findings: the current experiments suggest the sampling decision effect is not driven via a direct sampling-positivity association, but is malleable to people's understanding of their sampling behavior.

However, while much of the past research demonstrating the malleability of approach-avoidance effects has centered on people's interpretation of whether an action is approach, avoidance, or neither, recent theorizing has also argued that the valences of what approach and avoidance denote per se may also be malleable to interpretation (Mertens, Van Dessel, & De Houwer, 2018; Van Dessel et al., 2019). Such a possibility mirrors the findings in the current work, which manipulated sampling goals to change people's understanding of what their sampling behavior denotes. That is, the manipulation did not target the malleability of whether people interpreted an action as sampling or not, but rather, targeted the malleability of whether or not sampling behavior should serve as a positive signal for evaluative change. So, to the extent that there is overlap between sampling and approach behavior, the current results suggest it might similarly be possible to moderate approach-avoidance effects by changing people's interpretation of what approach and avoidance behavior mean for evaluation in a given context, rather than simply changing whether an action is interpreted as approach or avoidance in the first place. As such, the current work suggests value in future research exploring this additional step at which approach-avoidance effects may be malleable to interpretation.

6.3. Generalizability

The current experiments recruited samples of native English speakers online via Prolific.co, suggesting a predominantly Western pool of participants. However, samples recruited via that platform are generally more diverse than student samples recruited for lab studies at universities (see, for instance, the age range of the samples in the present experiments; Palan & Schitter, 2018; Peer, Brandimarte, Samat, & Acquisti, 2017). Additionally, there is good reason to expect relatively high generality of our findings inherent to the effects studied in the present work. First, the links between sampling or approaching positivity is considered an evolutionarily-basic proponent of survivability (Elliot, 2006). Second, people's flexible interpretations of their behavior, internal states, and what they denote have been shown to play a critical role in the impact of other effects across cultures (e.g., a belief in whether willpower is a limited or limitless resource; Savani & Job, 2017). Thus, there appears to be grounds to expect that the current effects—in which approach-avoidance effects are moderated by people's beliefs about what approach and avoidance denote—may also replicate cross-culturally. Of course, however, future research should test this possibility directly before making strong claims about cross-culture generalizability of the current effects.

6.4. Conclusion

Much of daily life is shaped by the autonomy we have over the stimuli we choose to interact with. Yet, much of the existing research on evaluative learning strips participants of this autonomy, leaving us with an understanding of evaluative learning that is mostly confined to passive learning settings. The current work builds upon previous recent evidence highlighting the value of supplementing findings from traditional passive learning approaches with those that provide participants autonomy in the learning process. We replicate an effect whereby choosing to sample a stimulus more frequently predicts a more positive evaluative shift, even if the stimulus is consistently paired with negative images. Critically, however, we find that this typical effect arises when people are sampling for positivity, but not when they are sampling negativity. In this way, the current experiments highlight the role that people's understanding of their sampling behavior's meaning has on their evaluations.

Open practices

We have taken various steps in the current work to meet the standards of best practices for scientific transparency and openness. Specifically, in the Methods section of each of the present experiments, we have reported how we determined our sample sizes, all data exclusions, and all of the manipulations and measures we used. Additionally, we pre-registered all the current experiments (Experiment 1: <https://aspredicted.org/5km3j.pdf>; Experiment 2: <https://aspredicted.org/5dr82.pdf>; Experiment 3a: <https://aspredicted.org/sq2a7.pdf>; Experiment 3b: <https://aspredicted.org/35sz6.pdf>; Experiment 4: <https://aspredicted.org/qf36w.pdf>). Finally, we have made the materials, data, and analysis scripts publicly available on OSF: <https://osf.io/3uzhk/>

We confirm that the authors have no personal financial interests that may have impacted the current manuscript. We also confirm that is not published or under consideration for publication at any other outlets. Additionally, as noted in the main text:

Declaration of Competing Interest

None.

Data availability

Links to the data, materials, and analysis scripts are shared in the paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jesp.2023.104516>.

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